

Smart Factories and Open Source Communities Must Work Hand-in-Hand

Many companies investing in open source development around IoT do not leverage open source communities. This article will help such companies understand the importance of communities and how they can help in a smart factory ecosystem.



A smart factory can self-optimize performance across a broad network, self-adapt and learn from new conditions in real or near-real-time, and autonomously run entire production processes. This indicates the quantum and complexity of software required to enable, and then better a smart factory. The best technology to make these software interfaces available can be none other than open source.

The term Internet of Things (IoT) encompasses a lot of verticals, including connected vehicles, smart transport, smart energy and resources, smart cities, digital government,

digital health, and many more. Digital factories, however, are the biggest revenue generating vertical among these. This vertical accounted for more than 12,000 million Euros worth of opportunity across the globe in 2020, which is forecast to grow to more than 28,000 million Euros by 2024.

A product manufactured in a smart or a conventional factory goes through various stages. These include planning, sourcing, development, production, order processing, customer service and relationship management. Each stage has a separate goal, which aligns with the goals of other stages to generate revenue for an organisation.

IoT weds open source

Each module of product development is connected to the others, and data generated in all the stages is vital for creating and improving products. The need to create a smart factory community is strongest in the production use case.

A typical manufacturing or shop floor consists of legacy systems, sensors, PLC, built-in production automation processes, SCADA, and a lot of other independent systems. Level 3 is usually the stage where data is gathered, and is referred to as the 'edge'. This data is either processed locally or sent to the cloud to create